

CLASS 13 / NOV 17 / U09

PARTS FOR THE LASER CUTTER

P4b CONTINUES — A CONTOUR BECOMES A PART YOU CAN HOLD

P4b CHECKPOINT

DISCRETIZING GEOMETRY

THICKNESS IS A DIMENSION

Last week a slice was a curve. Today it has a thickness, and the thickness is real.

INTERVAL = THICKNESS



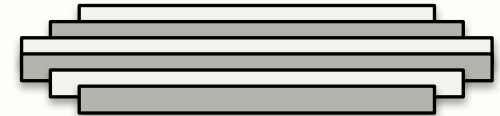
Parts touch. A solid stack with stepped edges.

INTERVAL > THICKNESS



Gaps between parts. Light passes; something must hold them.

INTERVAL < THICKNESS



Parts collide. Change the interval or the material.

MEASURE THE SHEET YOU WILL CUT. A NOMINAL THICKNESS IS NOT A MEASURED ONE — DESIGN TO THE NUMBER YOU MEASURED.

Slice count $\times$ thickness + gaps = assembled height. Check it against the 300 mm limit before you cut anything.

REGISTRATION: HOW PARTS FIND EACH OTHER

Twelve loose parts are not an assembly. Choose one system and draw it into every part.

SPINE

A cross member with notches. Each slice has a matching slot and drops onto it.

Order is fixed by the notches.

DOWEL

Two or more holes through every slice. Rods pass through and align the stack.

Holes must line up on every part.

SPACER

Separate small pieces set the gap between slices. Used with a dowel or glue.

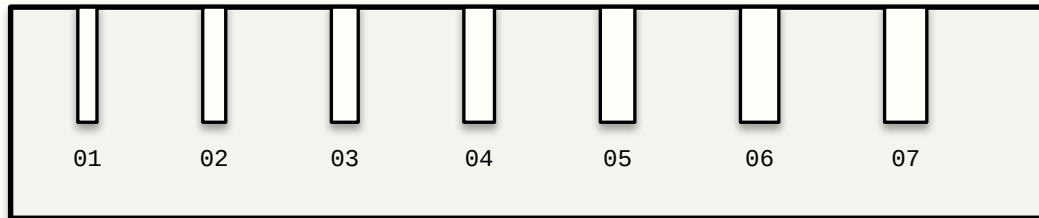
Spacers count as parts on the sheet.

THE REGISTRATION IS PART OF THE DESIGN. IT IS VISIBLE, SO DRAW IT ON THE PROCESS SHEET.

KERF, AND THE FIT TEST

The beam removes a thin strip of material. Every edge moves by half of it.

THE TEST COUPON



A strip of slots in steps of width, cut from your actual sheet. Push a scrap of the same sheet into each. The one that holds without forcing is your fit.

WHAT TO RECORD

- the measured sheet thickness
- the slot width that fit
- the kerf you measured
- the offset you applied: $\text{kerf} / 2$
- the machine settings the FabLab used

TEST BEFORE THE BATCH. THE COUPON AND ANY FAILED PARTS COUNT AGAINST YOUR ONE SHEET.

Coupon slot widths above are a diagram, not values to copy.

CLOSE AND CLEAN THE PROFILES

A contour from a mesh is rarely a clean part. Fix it before it becomes a cutting file.

CLOSED

Every part outline is one closed curve. An open curve is not a part.

SINGLE

No duplicate or overlapping lines. The beam cuts twice where lines overlap.

SIMPLE

Remove tiny fragments and islands you could not hold or glue.

SOUND

Check the thinnest neck of each part. If it would snap, change the profile.

CLEAN THE PROFILES IN THE GRAPH, NOT BY HAND IN THE CUTTING FILE — OR THE NEXT CHANGE UNDOES IT.

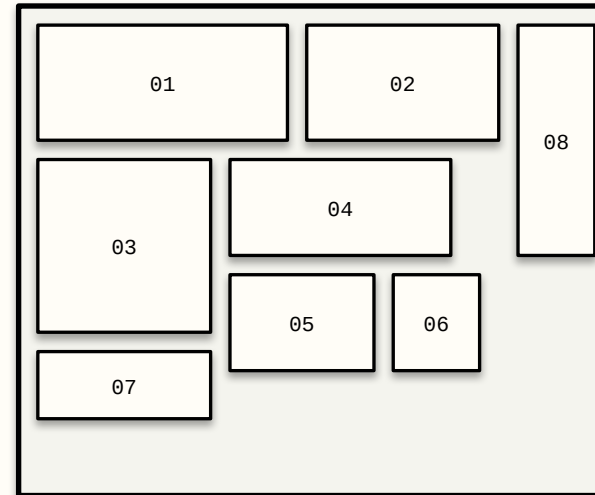
LABEL, ORDER, NEST

The slice order from last week becomes the part numbers. Then the parts go onto one sheet.

LABEL AND ORDER

- Number every part in slice order: 01, 02, 03 ...
- Engrave the number on a face the assembly will hide
- Add a mark for which way is up and which side faces front
- Keep a list: part, slice position, registration

NEST ON ONE SHEET



Keep a margin from the sheet edge

Leave a gap between parts

Reserve room for the coupon and one spare

P4b: FIT THE ASSEMBLY IN ONE CONFIRMED SHEET. IF IT DOES NOT NEST, CHANGE THE SLICES, NOT THE RULE.

THE CUTTING FILE, AND THE LAB

The machine reads lines, not intentions. Use the conventions the FabLab gives you.

FILE HYGIENE

- Vector lines only. Nothing is cut from an image.
- Cut and engrave lines kept apart, by the color or layer the lab specifies
- Real units, drawn at 1:1. Check one known dimension.
- Stroke weight set as the lab specifies
- Text converted to outlines

FABLAB PROCEDURES

- Safety orientation completed before you use a machine
- Book machine time; check what the lab has confirmed
- Confirm your material is approved for cutting
- Stay with the machine while it runs
- Follow staff instructions over anything on this slide

REGENERATE THE CUTTING FILE AFTER ANY GEOMETRY OR MATERIAL CHANGE. NEVER PATCH AN OLD ONE.

REMINDER — P4b

Visual documentation 20% · computational understanding 20% · technical execution 15% · fabrication quality 30% · experimentation 5% · requirements and identity 10%.

01

Use the selected P4a volume. Compare two section directions or intervals, then develop at least 12 ordered slices with material thickness and a registration system.

02

Fit the assembly within 300 mm on each axis and one confirmed material sheet. Test kerf and fit before the batch; label and nest every part.

03

Laser-cut and assemble the slices under FabLab procedures. Include test cuts and failed parts in the material allocation.

04

Present the assembly with one 17 x 11 inch process sheet showing the continuous volume, section sequence, cutting layout, assembly diagram and photographs.

05

Submit the editable source, cutting file and documentation. Regenerate fabrication files after any geometry or material change.

CHECKPOINT · WEEK 13 — Nov 17 — Review section directions, thickness, registration, fit/kerf test and cutting layout before production.

NOW — CHECKPOINT, THEN PARTS

Rest of the class: review with me first, then turn contours into parts.

✓

Section direction and interval chosen, with the comparison to show why

✓

Sheet thickness measured, and a registration system drawn into the parts

✓

A kerf and fit coupon drawn, ready for the lab

✓

Profiles closed and cleaned; every part numbered in slice order

✓

A first nesting layout on one sheet, with room for the coupon

NOTHING GOES TO THE CUTTER UNTIL THE COUPON FITS.

BEFORE NEXT CLASS

NEXT

Week 14 (target Nov 24) — Final production and assembly, photographing the work, and the booklet.

TEST

Cut the kerf and fit coupon under FabLab procedures. Record the fit and update the parts and the cutting file.

PREPARE

A production-ready cutting file: labeled, nested on one sheet, regenerated after the test.

TRACK

P4b is due at the final review (14% of the grade). Keep the test cuts and failed parts: they belong in the documentation.